

ABHILASH NEOG

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Research Interests

Scientific Foundation Models LLMs Multi-modal models Physics-Informed ML Time-Series Modeling

My research interests broadly lie in Scientific Foundation Models, Time-series modeling (sparse data, reasoning), LLMs (for continuous and non-textual data), Vision-Language Models (zero-shot evaluation, scientific applications), Physics-informed (or Knowledge-guided) Machine Learning.

Education

Virginia Tech

Ph.D., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0

Aug 2022 - May 2027

Blacksburg, USA

Virginia Tech

M.S., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0

Aug 2022 - Dec 2024

Blacksburg, USA

Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Engineering (B.E.), Computer Science, GPA: 8.08/10

July 2016 – July 2020

Pilani, India

Research Experience

KGML Lab, Virginia Tech | Graduate Research Assistant

Jan 2023 – Present

- Built a Model-agnostic Adapter for Irregular and Sparse Time-Series (TS) modeling
- Exploring LLM-guided knowledge-injection approaches into pretrained time-series foundation models via latent space editing
- Built a Scientific Foundation Model for spatio-temporal (probabilistic) predictions on irregular grids, leveraging multi-gpu training. Proposed an (a) *effective tokenization strategy* and (b) *Query-based forecasting*
- Developed a novel physics-guided Diffusion Model for Partial Differential Equation solving achieving *15-250x* faster inference
- Zero-shot evaluation of Vision-Language (VLMs) Models (*GPT-4*, *BLIP*, *LlaVA*, *Cog-VLM*, etc.) for *scene graph generation* and *reasoning ability* on VQA tasks involving scientific images

Publications

1. **Abhilash Neog**, Sepideh Fatemi, Medha Sawhney, et al. “LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data”. *KDD 2026*
2. **Abhilash Neog**, Arka Daw, Sepideh Fatemi, Medha Sawhney, et al. “Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling”. *TMLR, 2026* [pdf]
3. A. Pandey, **Abhilash Neog**, Gautam Jajoo. “On the Internal Semantics of Time-Series Foundation Models”. *BERT2S @ NeurIPS 2025* [pdf]
4. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Anuj Karpatne. “Beyond Loss Guidance: Using PDE Residuals as Spectral Attention in Diffusion Neural Operators”. *Under review (ML4PS @ NeurIPS 2025)* [pdf]
5. **Abhilash Neog**, Medha Sawhney, Sepideh Fatemi, et al. “Toward Scientific Foundation Models for Aquatic Ecosystems”. *FMSD @ ICML 2025* [pdf]
6. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Amartya Dutta, Arka Daw, Anuj Karpatne. “Physics-guided Diffusion Neural Operators for Solving Forward and Inverse PDEs”. *CV4Science @ CVPR 2025 (Oral)*
7. S. Fatemi, **Abhilash Neog**, Amartya Dutta, M. Sawhney, et al. “Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance”. *CV4Science @ CVPR 2025 (Oral)*
8. A. Dutta, M. Sawhney, K.S. Mehrab, **Abhilash Neog**, Mridul Khurana, et al. “Open World Scene Graph Generation using Vision Language Models”. *World Models @ ICML 2025* [pdf]
9. KS Mehrab, M. Maruf, Arka Daw, **Abhilash Neog**, HB Manogaran, et al. “Fish-Vista: A Multi-Purpose Dataset for Understanding Identification of Traits from Images”. *CVPR 2025* [pdf]
10. M. Maruf, Arka Daw, KS Mehrab, HB Manogaran, **Abhilash Neog**, M. Sawhney, et al. “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images”. *NeurIPS 2024* [pdf]
11. Baviskar, A., Ramanathan, K., **Abhilash, N.**, Pawar, D. and Bangalore, K., Oracle International Corp, 2024. “Machine Learning Based Spend Classification.” *U.S. Patent Application 17/903,161*. [pdf]
12. Lavika Goel, **Abhilash Neog**, Ashish Aman, and Arshveer Kaur. “Hybrid Nature-Inspired Optimization Techniques in Face Recognition” *Transactions on Computational Science XXXVI, Springer LNCS, 2020*. [pdf]

Industry Experience

Microsoft AI | Applied Scientist Intern

May 2026 – Aug 2026

- Image Generation/Editing, Bing Ads

Kryptowire | Machine Learning Intern

May 2023 – Aug 2023

- Developed an outlier detection model for denoising sensor-based Human Activity Recognition (HAR) **time series** data
- Built & deployed a CNN-based HAR model achieving 82% F-1 score on an android app using Keras & TensorFlow Lite

Oracle | Data Scientist

Sep 2020 – July 2022

- Built & deployed Machine Learning applications into ETL pipelines, leveraging **Spark** systems, **MLOps** & **CI/CD** pipelines
- Designed and deployed a *Demand Prediction* application for time series forecasting using the DeepAR model
- Developed an unsupervised classification algorithm achieving 40% higher accuracy than then **Language Models** on a 71k-label dataset.

VMware | Software Development Engineer Intern

Jan 2020 – June 2020

- Streamlined the process of fetching & filtering raw data from Workspace ONE Cloud using Spring Boot REST APIs
- Contributed to an end-user federation app on Workspace ONE Cloud, and wrote unit tests using JUnit and Mockito

Samsung Research Institute | Summer Intern

May 2019 – July 2019

- Performed a feasibility study of Multi-frame Noise Reduction solutions' deployment in Live Focus for Low light conditions
- Optimized the existing HAL call flow, in C++, with considerable noise reduction in the first phase of live focus capture

Selected Projects

Can Large Vision Language Models (VLMs) Ground Fine-grained Attribute? pdf Aug '24 – Dec '24

- Developed a novel dual-scale attention framework for fine-grained attribute localization in Large Vision-Language Models (**LLaVa**), incorporating entropy-based head selection, maximally connected component filtering, and hierarchical constraints

Evaluating Model Reasoning & Hallucinations in Medical LLMs code pdf Jan '24 – April '24

- Analyzed and evaluated factual error propagation in open-source medical LLMs such as BioMistral, Asclepius, Alpacare, and PMC-LLaMA to identify variations in their efficacy and ensure reliable information dissemination in medical settings.

Convergence analysis of PINN for solving inverse PDEs code pdf Aug '23 – Dec '23

- Performed adaptive weighing of physics-based and data-driven loss terms in Physics-informed Neural Networks
- Achieved 50% average error reduction in PDE (Partial Differential Eq.) parameter estimation of Burgers & Allen-Cahn eq.

Mathematical Reasoning in Large Language Models (LLMs) code pdf Aug '23 – Dec '23

- Worked on the problem of numerical headline generation and numeral masked-fill as part of NumEval @ SemEval 2024
- Adapted **Llama**, **T5**, **BART** & **RoBERTa** models by Direct **fine-tuning** & **prompt engineering** for the respective tasks

Text Summarization of Electronic Theses and Dissertations (ETD) pdf Sept '22 – Dec '22

- Developed a text summarization pipeline, integrating both Transformer-based abstractive algorithms (pre-trained Pegasus & RoBERTa) and traditional extractive algorithms like TextRank, LexRank & LSA, within an ETD Info. Retrieval system

Technical Skills

Languages: Python, Java, C++, SQL, R

Frameworks: PyTorch, Git, Hydra, Tensorflow Keras, Spark

Miscellaneous

• Reviewer:

- Conferences/Journal - NeurIPS 2025, AAAI 2026, TMLR, ICML 2026, KDD 2026, ECCV 2026
- Workshops - FM4Science (ICLR '26), FMSD (ICML '25, '26), FMs in the Wild (ICLR '25), Open Science for FMs (ICLR '25), XAI4Science (ICLR '25), ML4PS (NeurIPS '25), TSALM (NeurIPS '24)

• Talks:

- Invited talk at the **ASLO 2026 Conference - Session on AI for the Next Frontier in Aquatic Sciences**
- Invited talk at the **AGU 2025 Conference - Session on Advancing Water Science Through AI**
- Talk on Transfer Learning in Lake Ecosystems at the “*NSF Macrosystems Biology Meeting 2024*”
- Lightning Talk at the “*Frontiers in Ecological Forecasting 2023*,” event at Virginia Tech

• Awards:

- **NSF NAIRR (National AI Research Resource) Pilot Award, 2024**
- “*Star of the Month (Dec 2021)*” within the Oracle Analytics Cloud Organization, Oracle India

• Teaching:

- Graduate Teaching Assistant, CS 5805 Machine Learning, Spring 2024
- Teaching Assistant, BITS F312 Neural Networks and Fuzzy Logic, Fall 2019